



Adaptive Budget Allocation for Online Advertising

Learning to bid under a hard budget as campaign scale and market dynamics evolve

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A controlled study of learning, budget pacing, and adaptation

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PROBLEM DEFINITION

Auction Model and Budget Constraint

First-price auction, each round

$$\text{win} \iff b_{i,t} \geq m_{i,t}$$

The highest competing bid for campaign i is the only uncertain quantity in the auction.

Utility and cost on a win

$$f = v_i - b_{i,t}, \quad c = b_{i,t}$$

Higher bids win more often but earn less per win and spend more budget.

Global budget constraint

$$\sum_t c_t \leq B, \quad \rho = B/T$$

The initial spend rate is $\rho = B/T$. Effective learners must reconcile it with the remaining budget.

CONFLICT GRAPH

With multiple campaigns, the learner selects a super-action: one feasible bid per campaign subject to the conflict graph.

The investigation expands in scope:

1 to 4 campaigns

stochastic to changing markets

semi-bandit and full feedback

Failure Diagnosis and Controlled Improvement

STEP 1

Reference learner

Start from established methods: UCB, linear programs over distributions, Hedge, and online gradient descent.

STEP 2

Failure diagnosis

Track regret, expenditure, selected bids, and policy evolution to identify the binding limitation.

STEP 3

Targeted intervention

Change only the component linked to the failure while keeping the environment and benchmark fixed.

STEP 4

Controlled ablation

Run the same environments and seeds while changing one component at a time.

Every comparison is paired. Baseline and revised learners see the same environment realizations, so the curves isolate algorithmic differences.

Single Campaign Under a Hard Budget



Two reference learners: budget-blind UCB1 and a budget-aware UCB learner whose action distribution comes from a linear program.

$$T = 5000$$

$$B = 125 \rightarrow \rho = 0.025$$

$$v = 0.6$$

$$3 \text{ uniform competitors} \rightarrow P(\text{win}) = b^3$$

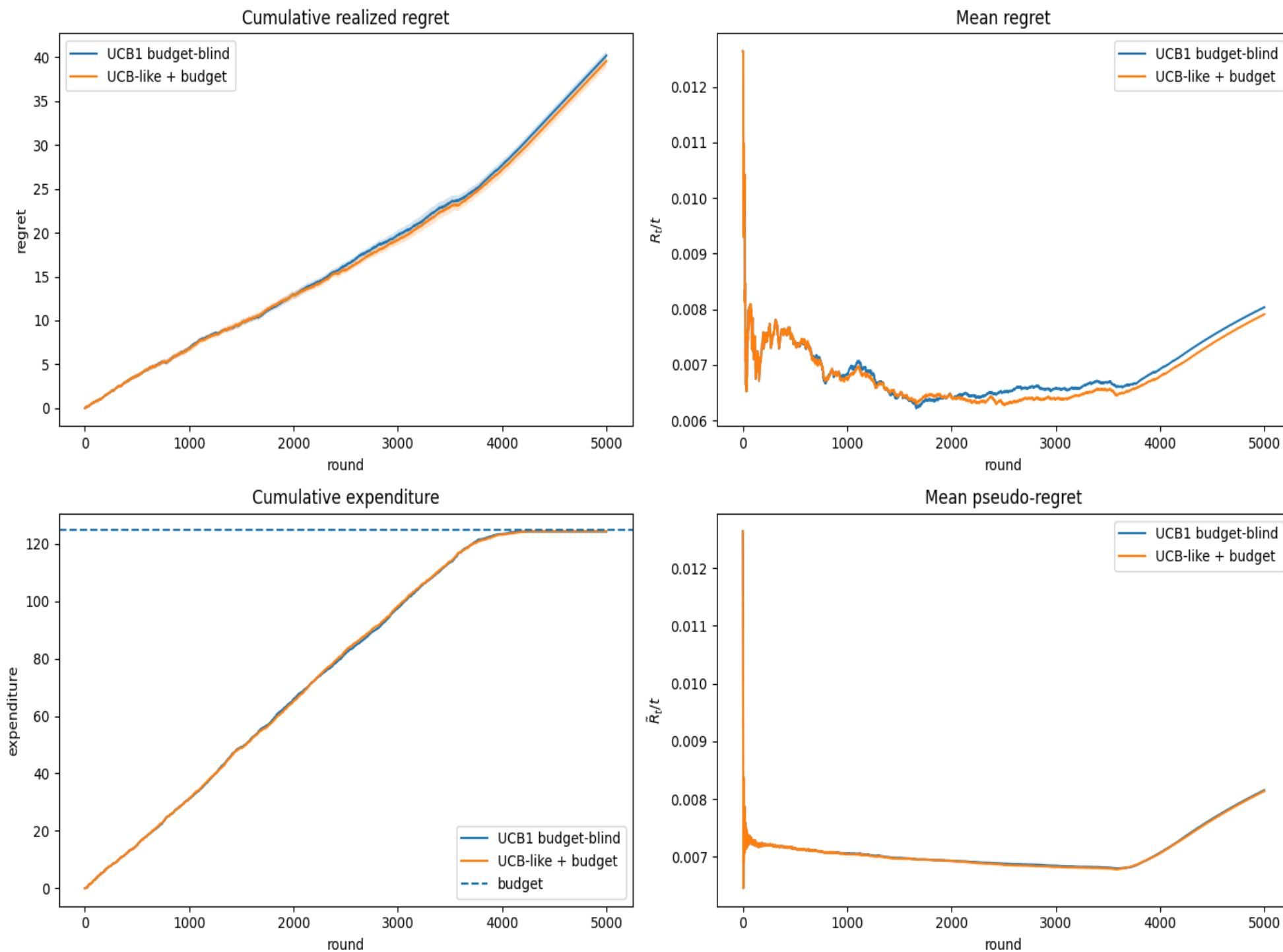
$$\text{discrete bid grid, } b \leq v$$

BENCHMARK

Stochastic clairvoyant: LP over bid distributions with true win probabilities, spend $\leq \rho$. On this instance it mixes 96.6% on bid 0.4 — expected reward ≈ 0.0126 per round.

STAGE 1: FAILURE DIAGNOSIS

Wide Confidence Intervals Neutralize the Budget Constraint



40.2

final regret, UCB1
budget-blind

39.6

final regret, UCB-like +
budget LP

Why the budget-aware learner behaves like UCB1

Hoeffding intervals stay wide for most of the horizon.

The cost LCB often equals zero, so the LP treats many bids as nearly free.

Both learners exhaust the budget early, play the null bid in about one third of the rounds, and obtain about 36% of the clairvoyant reward.

Budget estimation is the binding limitation.

A Bernoulli Model Makes Rare Wins Learnable

1 WIN PROBABILITY

$$\hat{r} = (v - b) \hat{p}, \quad \hat{c} = b \hat{p}$$

At a fixed bid, one Bernoulli outcome determines both reward and cost. We estimate its probability and derive both quantities from it.

2 KL CONFIDENCE BOUND

$$U_t = \max\{q \in [\hat{p}, 1] : N \text{kl}(\hat{p}, q) \leq \log t\}$$

KL bounds replace Hoeffding bounds and tighten faster when winning probabilities are small.

3 AFFORDABILITY FILTER

$$b \text{ playable} \iff b \leq B_t$$

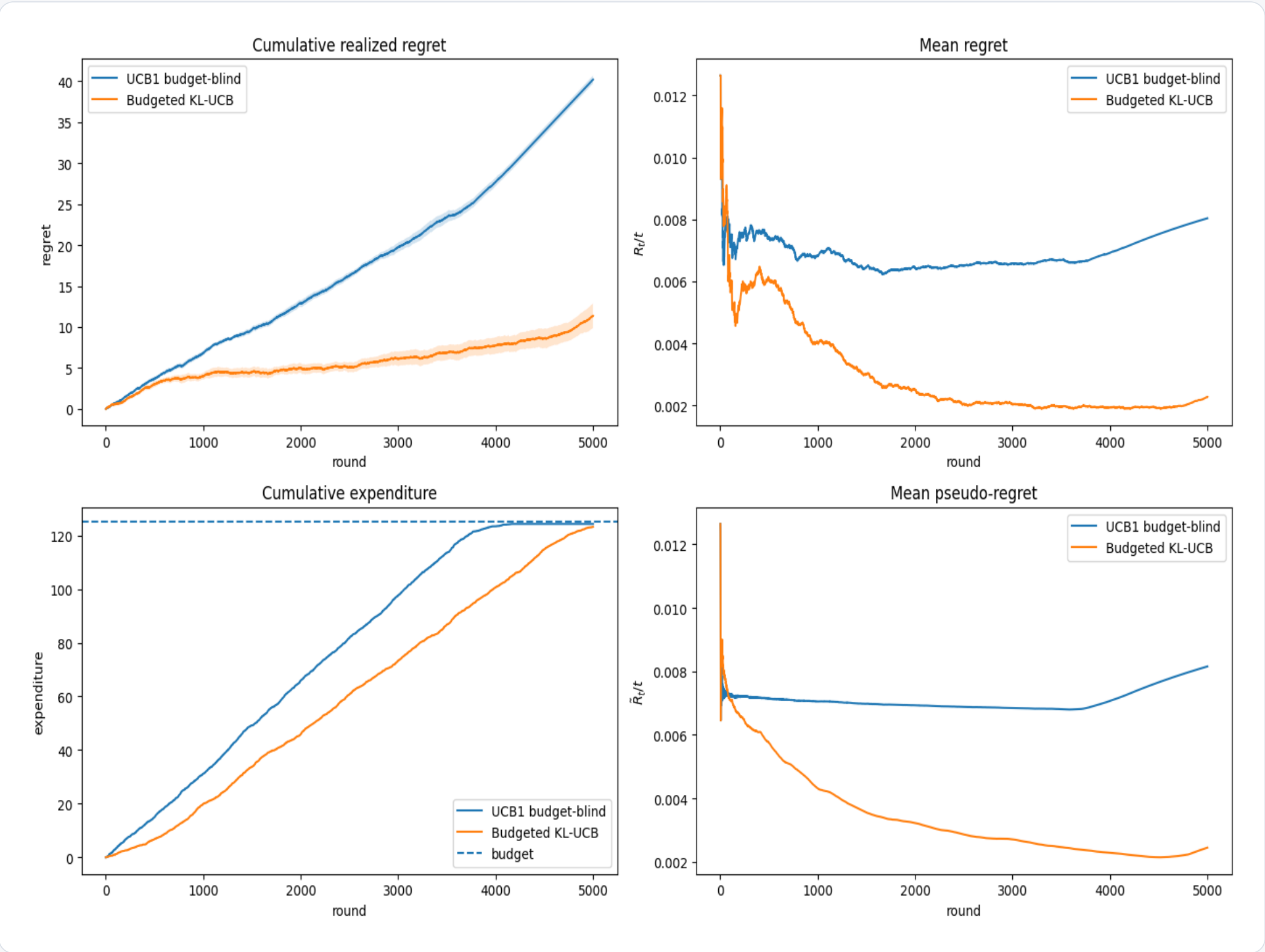
The LP considers a bid only when the remaining budget can pay for it. Bid 0 remains the zero-cost action.

BERNOULLI KL DISTANCE $\text{kl}(p, q) = p \ln \frac{p}{q} + (1 - p) \ln \frac{1 - p}{1 - q}$ KL measures the discrepancy between two Bernoulli rates. Its geometry yields tighter upper bounds near zero and one.

WHY KL FITS THIS AUCTION Relevant win probabilities are small: $P(\text{win}) = b^3$ is 0.027 at $b = 0.3$ and 0.064 at $b = 0.4$. Hoeffding optimism saturates and can rank bids in the wrong order. The Bernoulli KL interval concentrates faster near zero.

STAGE 1: EVIDENCE

KL Confidence Bounds Reduce Final Regret to 11.4



40.2 → 11.4 final regret

82%

of clairvoyant reward
(was 36%)

123 / 125

budget used while
preserving affordability

ABLATION WITH MATCHED SEEDS

KL: regret **11.4** and spend 123/125. Hoeffding: regret **35.7** and spend 45/125. Loose optimism keeps selecting low bids that rarely win.

Shared Budget Across Multiple Campaigns



Each round selects one feasible bid profile over the conflict graph. Semi-bandit feedback reveals outcomes for selected campaign-bid pairs.

4 campaigns · $v = (0.95, 0.85, 0.78, 0.90)$

$T = 3000$ · $B = 700 \rightarrow \rho \approx 0.233$

conflict chain 0-1-2-3

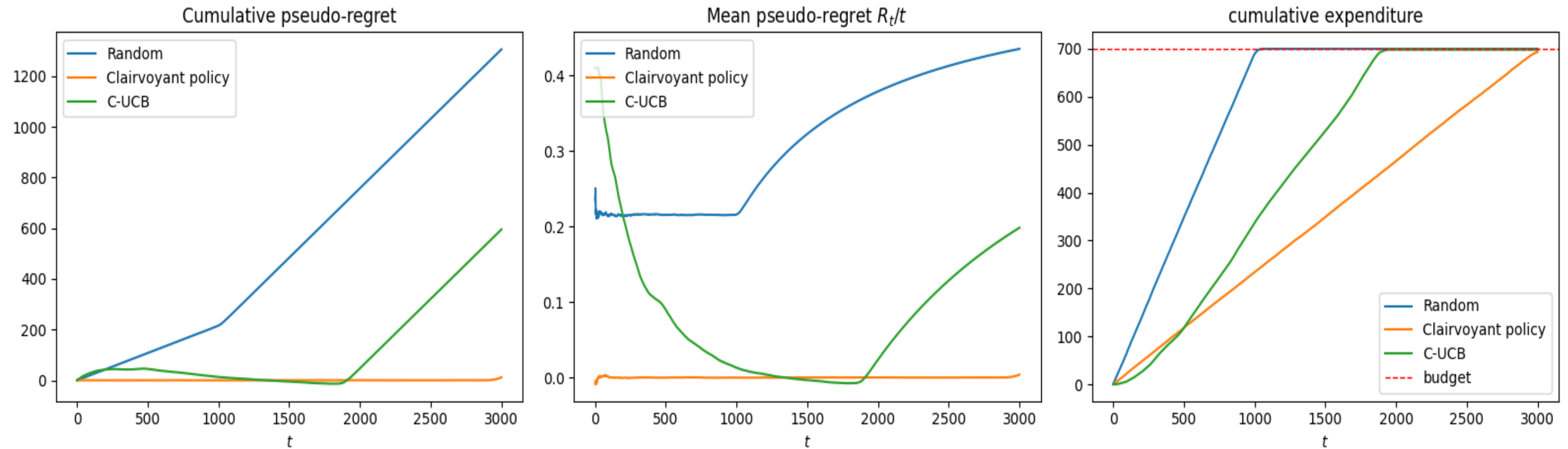
33 atomic arms · 250 super-actions

Beta marginals + 30% common factor $\rightarrow$ corr ≈ 0.3

BENCHMARK Clairvoyant LP over feasible super-actions with true expected reward and cost under the average budget rate $\rho = B/T$.

STAGE 2: FAILURE DIAGNOSIS

The Learner Finds Good Actions but Spends Too Early



MID-HORIZON

Pseudo-regret stays near zero. The action estimates are already accurate.

$t \approx 1900$

Cost lower bounds understate expenditure. The learner exhausts the budget near round 1900 and then selects only the null action.

FINAL

Final pseudo-regret reaches about 595, compared with 0.435 per round for the random policy. Maximum spend remains 699/700.

Empirical Costs Align Spending with the Horizon

1 WIN PROBABILITY PER ARM

$$\hat{r} = (v_i - b) \hat{p}, \quad \hat{c} = b \hat{p}$$

One Bernoulli estimate for each campaign and bid. Semi-bandit feedback supports an independent update for every selected arm.

2 KL BOUND FOR WIN PROBABILITY

$$\max\{ q \geq \hat{p} : N \text{kl}(\hat{p}, q) \leq \log t \}$$

KL bounds improve the ranking of campaign-bid pairs with small winning probabilities.

3 EMPIRICAL COST IN THE LP

$$\hat{c}(a) = b \hat{p} \quad (\text{not the LCB})$$

The LP uses the unbiased empirical cost estimate. The ablation isolates its effect on early expenditure.

4 DYNAMIC PACING

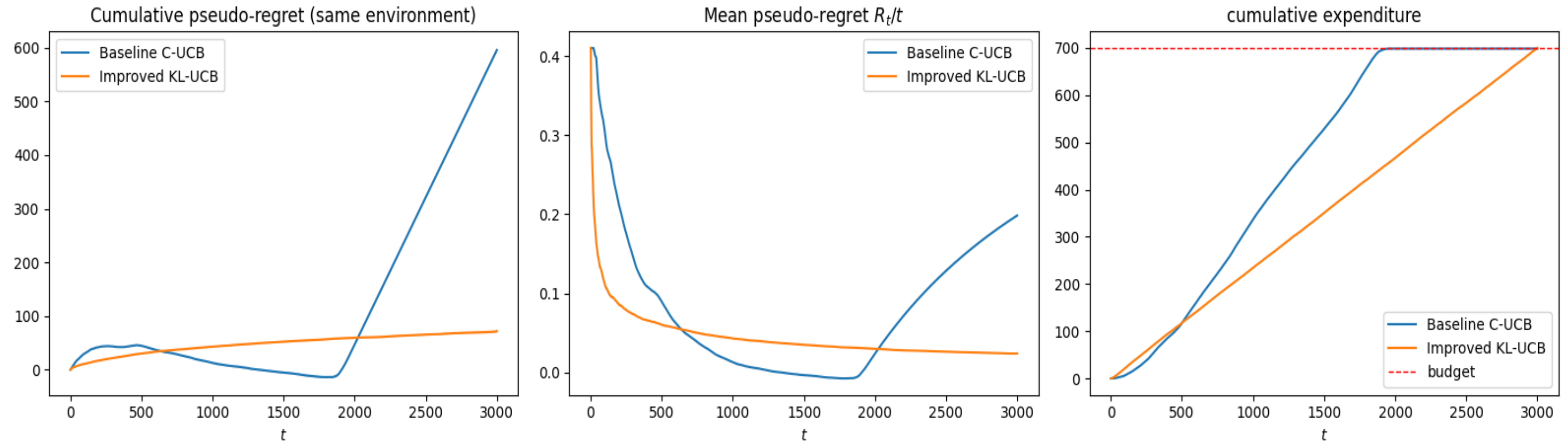
$$\rho_t = B_t / (T - t)$$

The spend cap becomes remaining budget divided by remaining rounds. The LP also excludes unaffordable super-actions.

The linear program remains unchanged. Only its statistical estimates and budget cap change.

STAGE 2: EVIDENCE

Dynamic Pacing Extends Strong Performance to the Horizon



$\approx 595 \rightarrow \approx 72$

final pseudo-regret with matched seeds

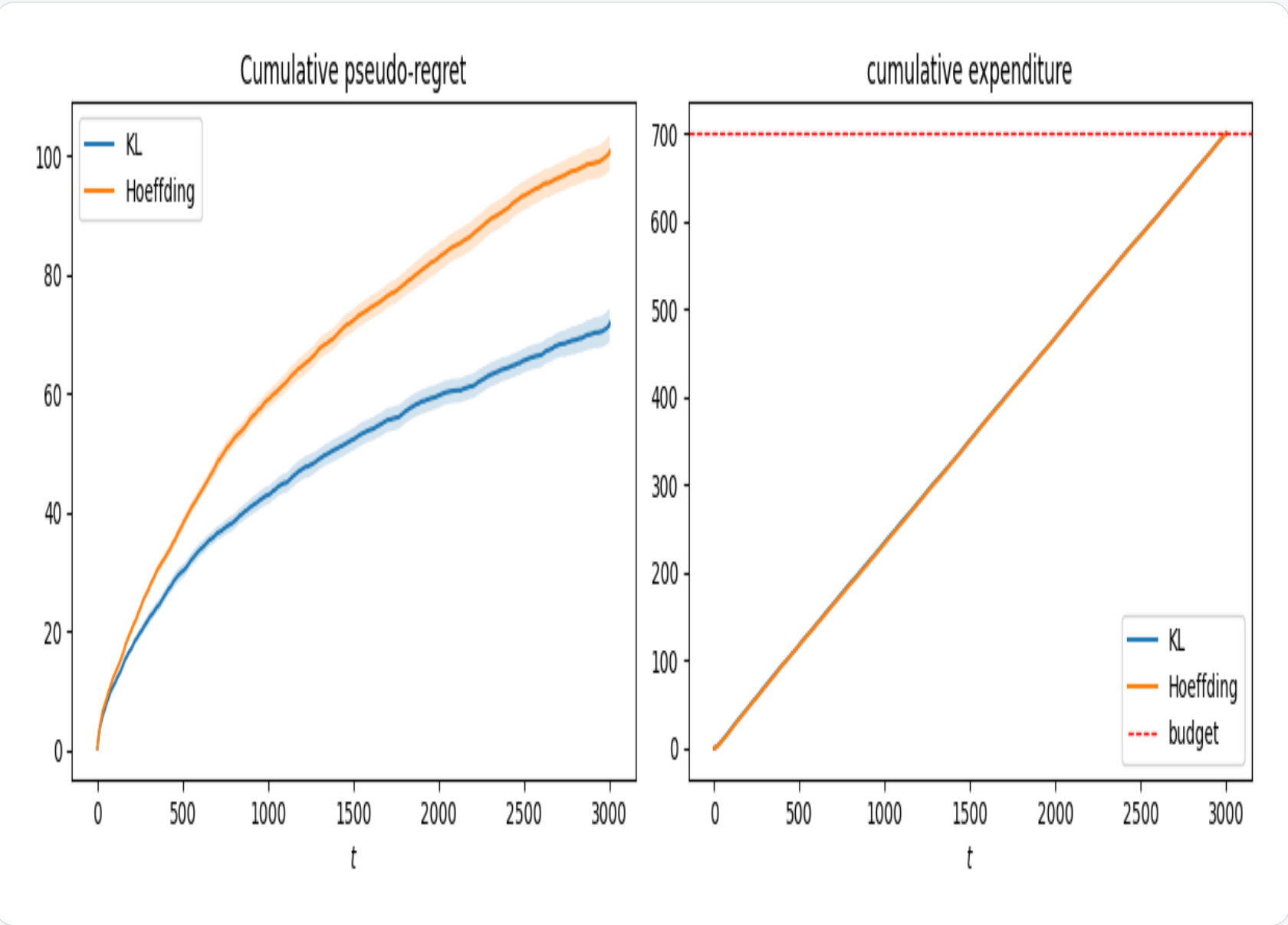
96%

of the oracle's expected reward

0.024

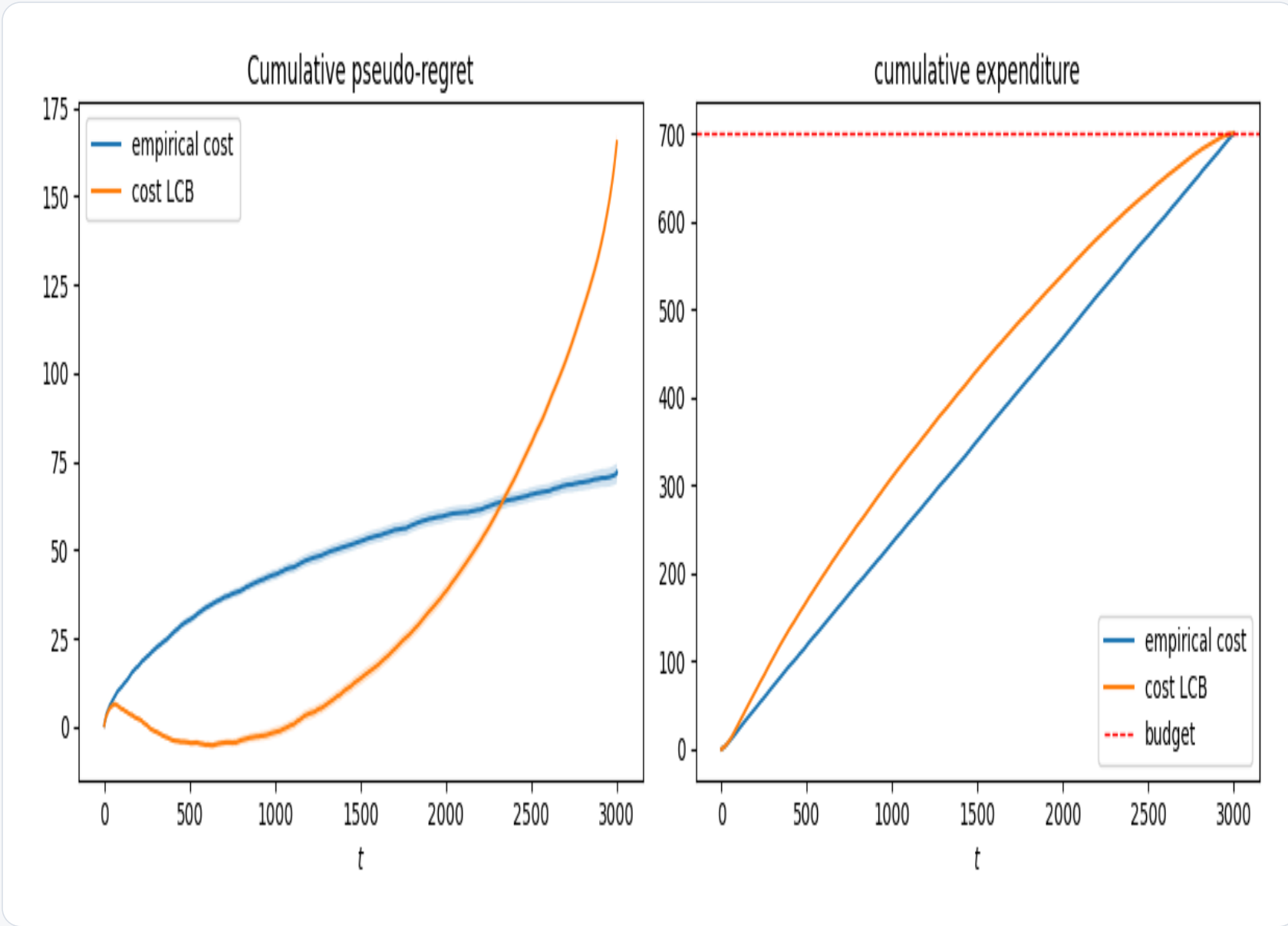
regret per round at the horizon, still decreasing

Confidence and Cost Estimates Produce Separate Gains



72 vs
101

KL versus Hoeffding. Only the confidence bound changes, and KL ranks small-probability arms more accurately.

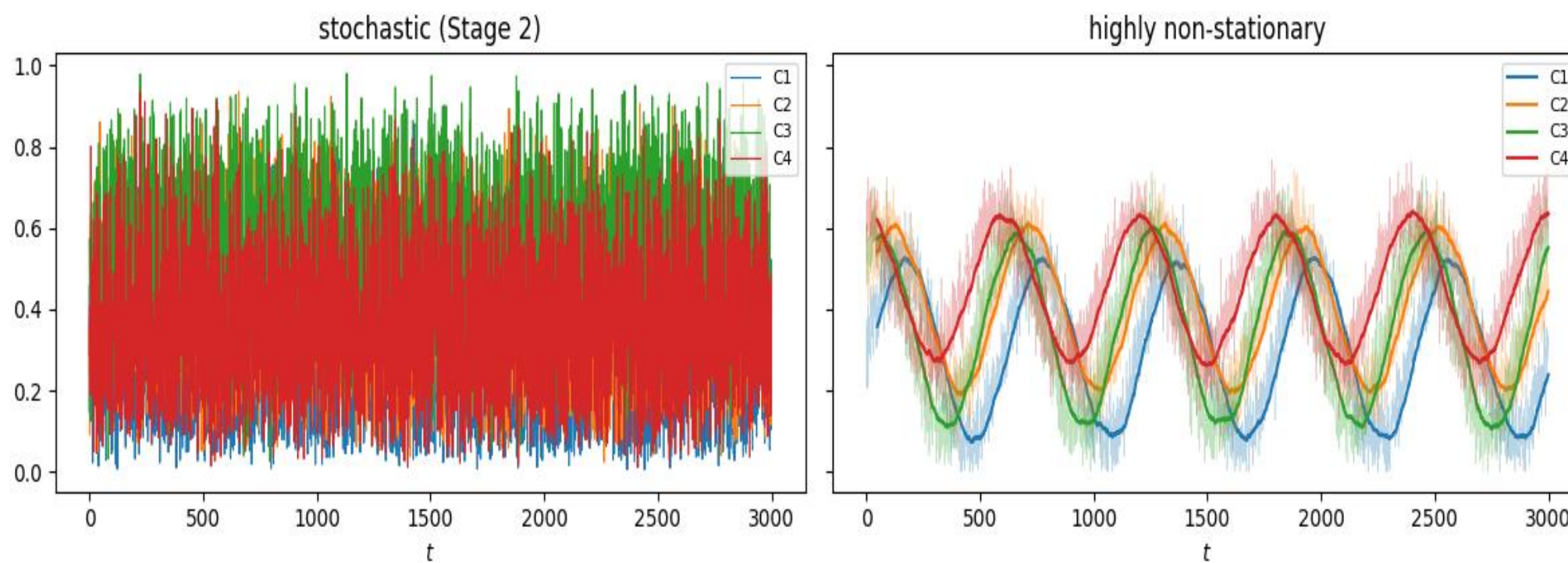


72 vs
165

Empirical cost versus cost LCB. The LCB spends 241 by $T/4$ instead of 177, and its early advantage disappears before the horizon.

Primal-Dual Budget Control

The same combinatorial bidding instance now provides full feedback. After each round, thresholds reveal reward and cost for every super-action.



Stochastic: Beta marginals + common factor

Highly non-stationary: oblivious fast sinusoids + small noise

$T = 3000 \cdot B = 700 \cdot 250$ super-actions

Hard budget enforcement

BENCHMARK

Best fixed distribution in hindsight under the expected budget constraint. We compare against full OPT, a stricter target than ρ times OPT in the adversarial guarantee.

Conservative Normalization Leaves Budget Unused

PRIMAL LEARNER: HEDGE ON 250 SUPER-ACTIONS

$$L_t(a) = f_t(a) - \lambda_t (c_t(a) - \rho)$$

DUAL LEARNER: PROJECTED OGD, $\lambda \in [0, 1/\rho]$

$$\lambda_{t+1} = \Pi(\lambda_t + \eta (c_t \cdot x_t - \rho))$$

REFERENCE LEARNING RATES

$$\eta_H = \sqrt{\log K/T}, \quad \eta_{\text{OGD}} = 1/\sqrt{T}$$

0.249 / 0.222

regret/T — stochastic / non-stationary (random: 0.436 / 0.391)

569 / 558

spend out of 700 with no violations

DIAGNOSIS

The learner beats the random policy in both regimes, but wide normalization makes its updates small and leaves budget unused.

Hedge needs losses in [0,1], but the Lagrangian contains a changing dual variable. Fixed worst-case normalization creates a range near 11 and reduces the effective update size.

Calibrated Learning Rates Improve Budget Use

1 RATES CALIBRATED TO THE UPDATE RANGE

$$g_t(a) = f_t(a) - \lambda_t c_t(a)$$

$$\eta_H = \sqrt{8 \cdot \log K / T}, \quad \eta_{\text{OGD}} = \lambda_{\max} / (G\sqrt{T})$$

Hedge updates only on the action-dependent part of the Lagrangian. Learning rates follow the theoretical bounds for the resulting Hedge and OGD ranges.

2 PER-ACTION BUDGET CHECK

a_t unaffordable $\Rightarrow$ play null action

Only an unaffordable sampled action is replaced with the null action.
Affordable actions remain available.

Declared asymmetry: Hedge and the dual still update on the proposed x_t , while the budget decreases with the cost of the executed action.
Override frequency is ≈ 0 in both regimes.

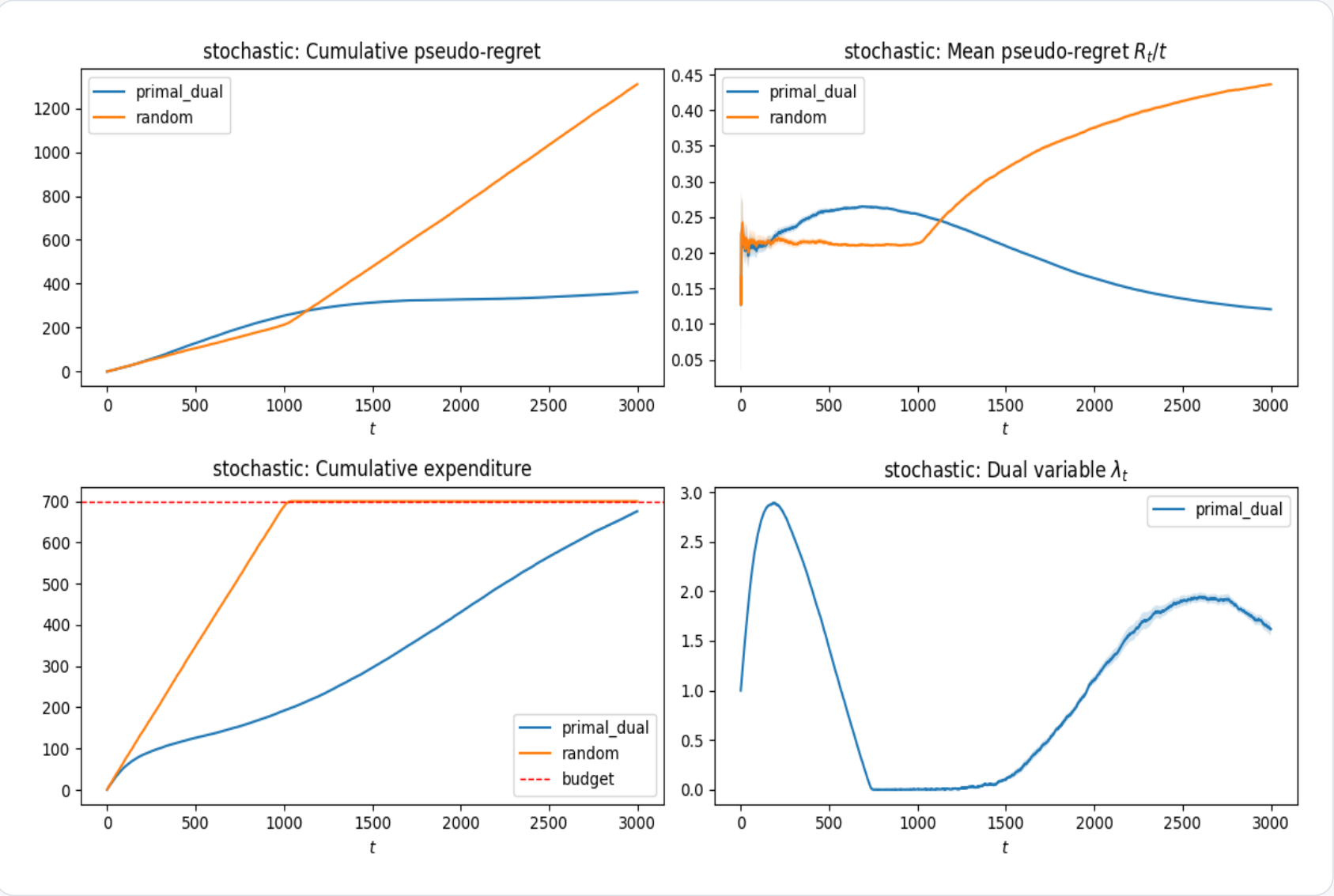
DUAL VARIABLE AS PACING

The dual variable already prices overspending. It rises when average expenditure exceeds p and reduces the appeal of costly actions.

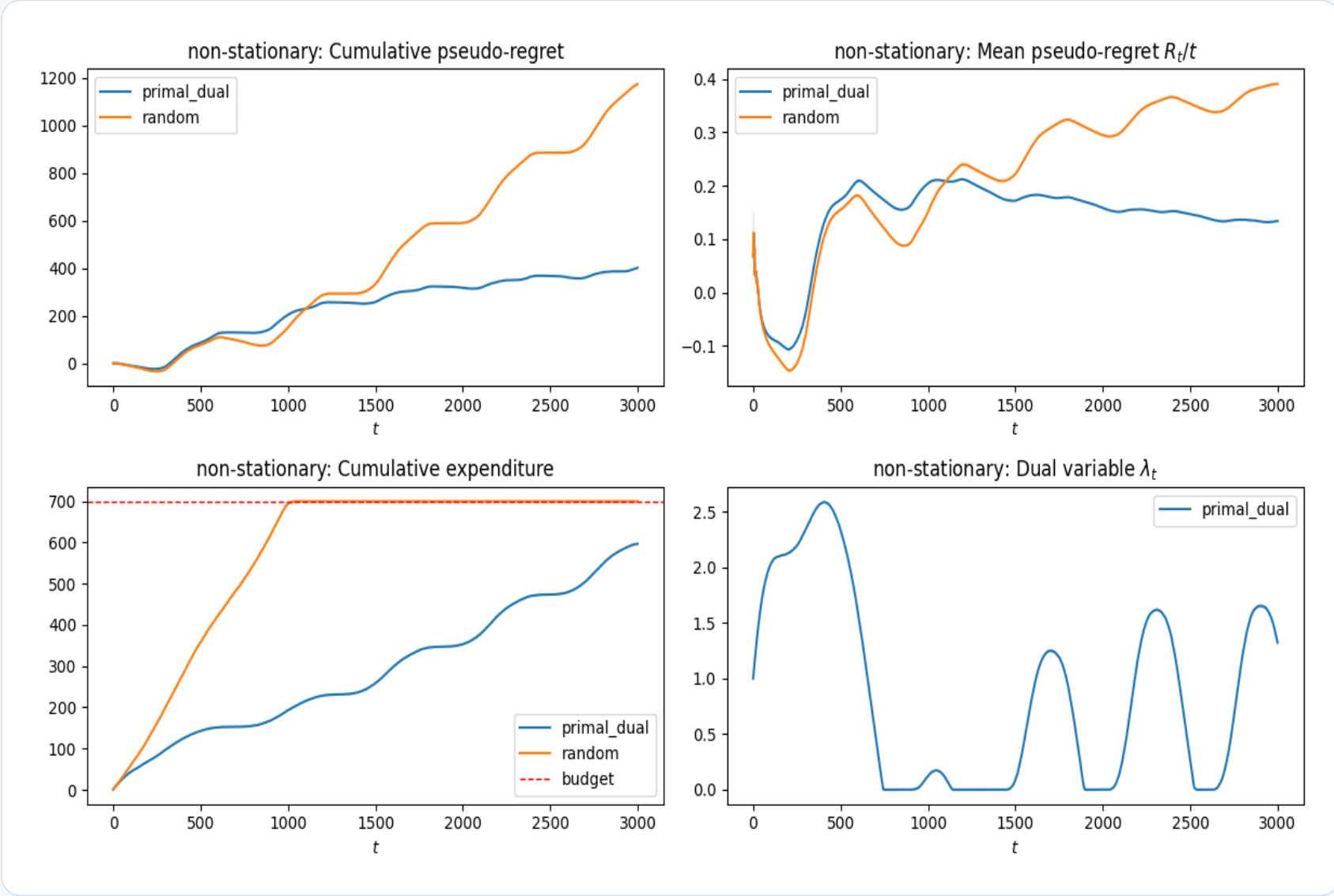
The ablation attributes the observed gain to calibrated learning rates. The per-action check provides a cleaner safety rule but has negligible empirical effect on this instance.

Calibrated Rates Improve Both Regimes

STOCHASTIC



HIGHLY NON-STATIONARY



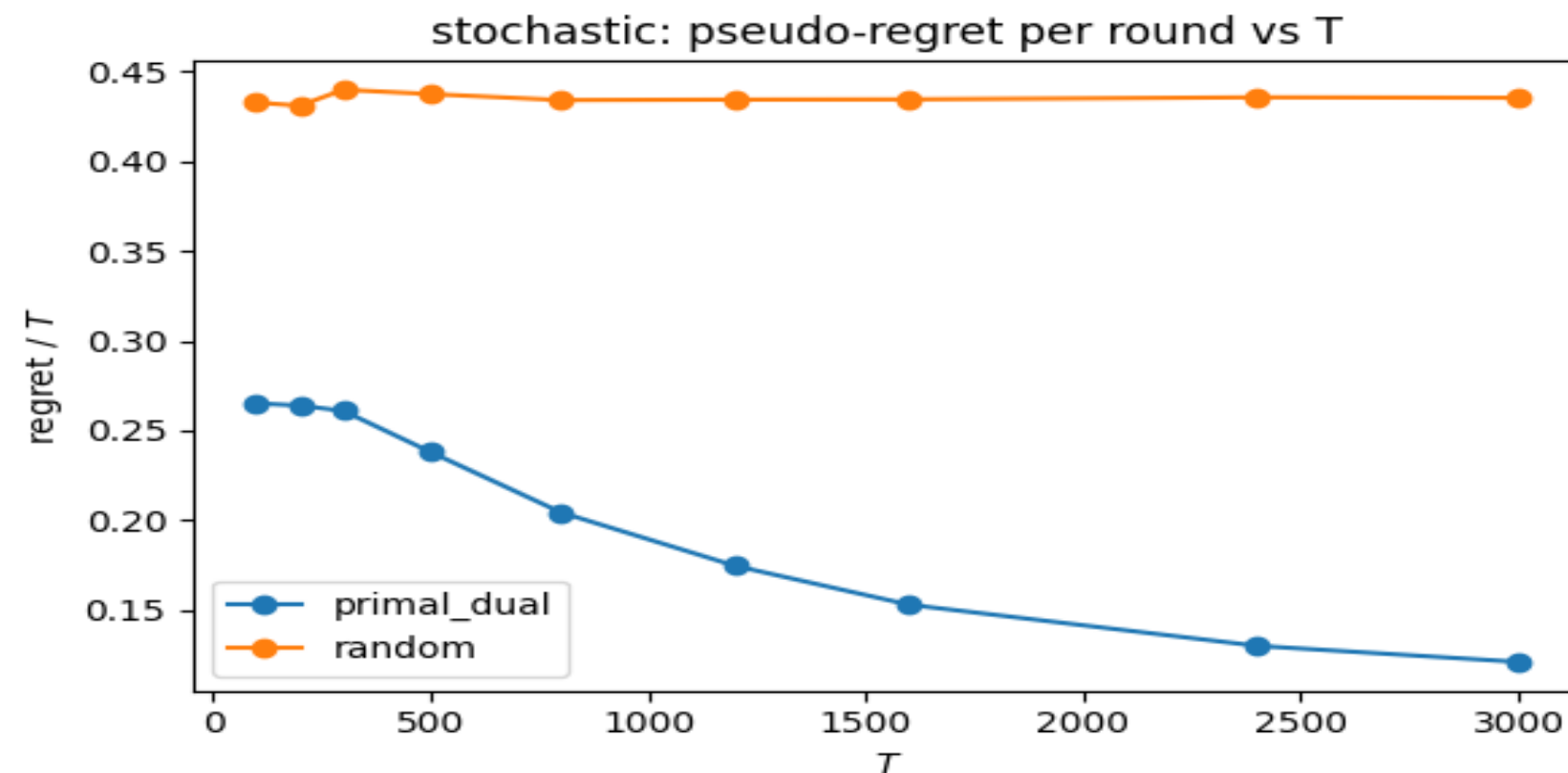
0.249 → 0.121 regret/T,
stochastic

0.222 → 0.134 regret/T, non-
stationary

569→676 · 558→597
spend out of 700 with no violation

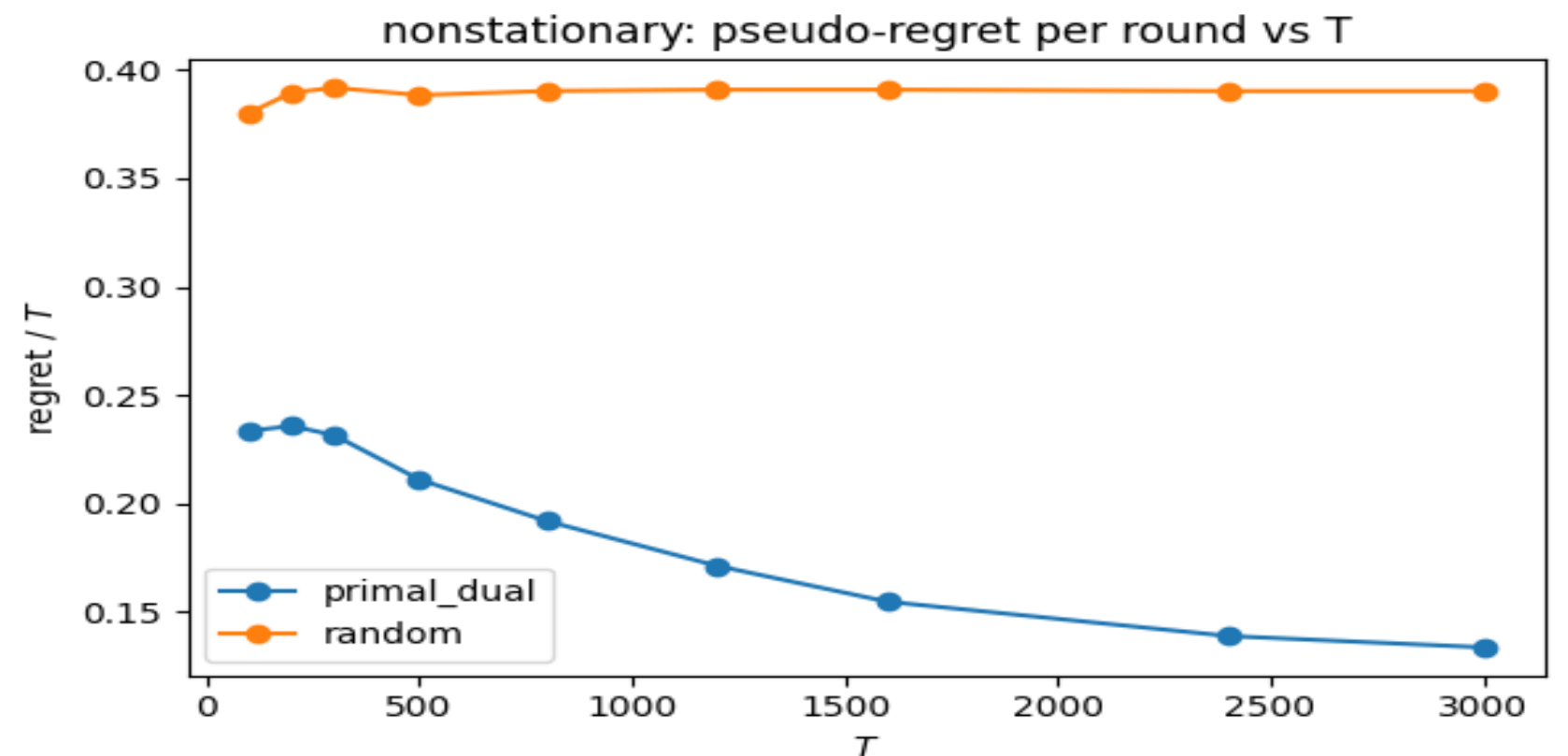
STAGE 3: ROBUSTNESS

Regret per Round Decreases with the Horizon



0.27 → 0.12

regret/ T as T grows, stochastic — consistent with sublinear regret



0.23 → 0.13

regret/ T as T grows, non-stationary — same algorithm, same constants

Across horizons from 100 to 3000, primal-dual regret per round decreases in both environments while the random policy stays high and nearly flat.

Budgeted Learning in a Changing Auction Market



The combinatorial learners now face three market regimes. Sliding windows and change detection compete with the full-feedback primal-dual method as profitable campaigns change over time.

HORIZON

$T = 3000$

3 equal segments, $L = 1000$, 2 known change points ($\gamma = 2$).

BUDGET

$B = 560$

First-price auctions, 4 campaigns, $\rho = B/T \approx 0.187$.

ALGORITHMS

SW Combinatorial-UCB

CUSUM Combinatorial-UCB

Primal-dual (full feedback)

BENCHMARK

≈ 1229

segment-local oracle: one LP per segment, same p per round.

STAGE 4: MARKET STRUCTURE

Piecewise-Stationary Market

T = 3000 · 3 segments · B = 560

One bid per campaign each round, from a **grid of 8 values** , denser on low bids (step 0.10, then 0.15).

Conflict chain C1–C2–C3–C4: adjacent campaigns cannot both bid → **168 super-actions**, 27 positive arms.

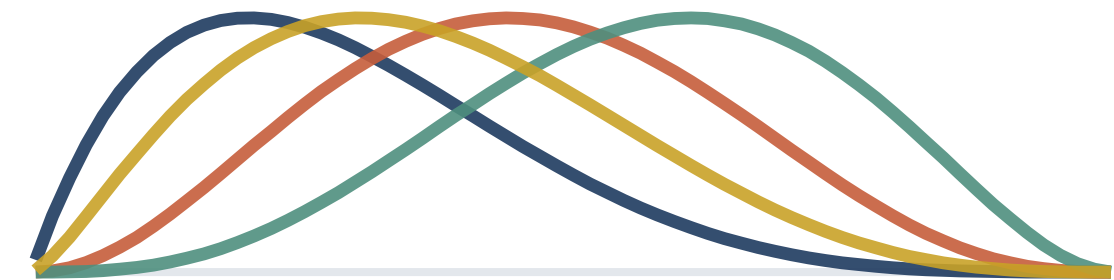
Each campaign has a hidden threshold $m_i \sim \text{Beta}(\alpha_{s,i}, \beta_{s,i})$, fixed within a segment, new at each change point.

Win where $b_i \geq m_i$, pay the bid (first price): utility $v_i - b_i$, cost b_i .

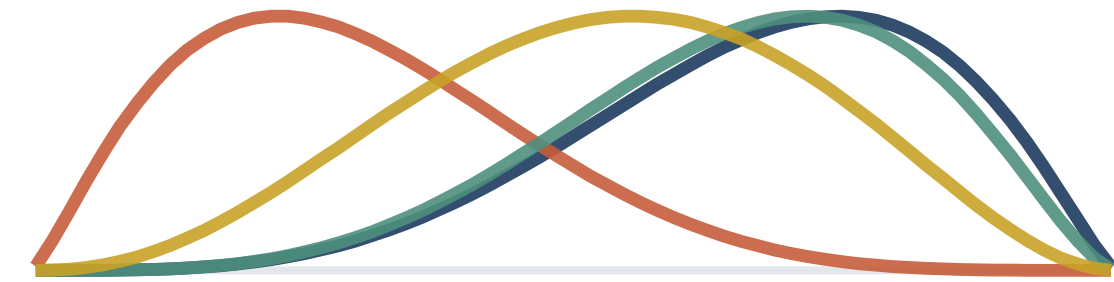
All campaigns draw from a **shared budget** B = 560, $\rho = B/T \approx 0.187$ per round.

The cheapest campaign rotates at every change point:

SEGMENT 1
t = 1-1000
cheapest: **C1**



SEGMENT 2
t = 1001-2000
cheapest: **C2**



SEGMENT 3
t = 2001-3000
cheapest: **C3**



distributions of m_i · x: 0 to 1



Three Ways to Handle Stale Information

Hyperparameters follow analytical formulas. The number of change points is known.

FORGETS BY CONSTRUCTION

SW-UCB



- cache of the last W rounds only
- bonus computed with $\log W$
- arm unseen in window: one forced pull

$$W = 2\sqrt{T \log T / \Upsilon_T} \approx 219$$

DETECTS, THEN RESETS

CUSUM-UCB



- M reference samples per arm $\rightarrow$ mean
- statistic accumulates deviations; over $h \rightarrow$ arm reset
- α -exploration keeps arms monitored

$$M = 7, h \approx 14.6, \alpha \approx 0.07$$

NEVER FORGETS

Primal-dual

$$L(a) = f(a) - \lambda (c(a) - \rho)$$

- Hedge on $K = 168$ super-actions, full feedback
- OGD on the dual λ (budget price)
- gains normalized in $[0,1]$, per-action budget stop

$$\eta_{\text{Hedge}} \approx 0.041, \eta_{\text{OGD}} \approx 0.018$$

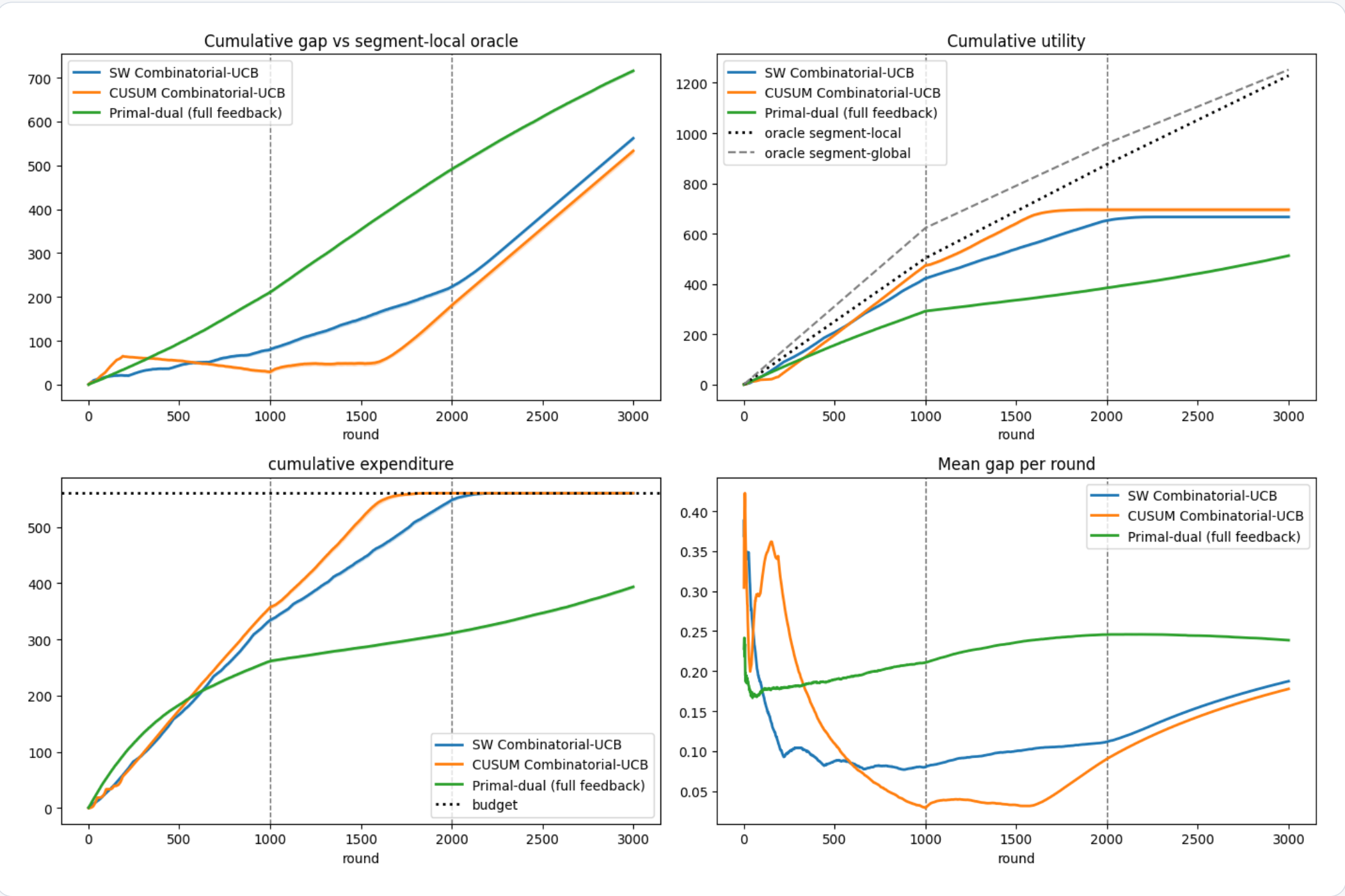
SHARED DECISION LAYER

Both UCB learners send reward UCBs and cost LCBs to an LP over feasible super-actions with a fixed cap $\rho = B/T$.

STAGE 4: FAILURE DIAGNOSIS

Early Spending Limits Relearning After Change Points

30 trials · final gap vs segment-local oracle ≈ 1229



PROBLEMS TO FIX

- ① both UCBs spend all of B, but **too early** : cost LCB understates costs, fixed p never corrects
- ② primal-dual leaves **167 of 560 unused** : infinite memory, conservative
- ③ CUSUM fires **~18 resets vs 54 reference** : too few samples per arm to reach $h \approx 14.6$ (no hand-tuning of h)

Mean over 30 trials, shaded SEM · dashed lines: change points · dotted: oracles and budget.

S4: INTERVENTION

Memory and Pacing Preserve Future Capacity

Same environments, benchmarks, and seeds

01 · WIDER WINDOW

$$W = L/2 = 500$$

- half a segment: widest window that stays comfortably stationary
- more samples per arm in the combinatorial setting
- derived from known quantities, **not tuning**

ablation, 30 trials each: gap **496 → 417** vs $W = 219$

02 · EMPIRICAL COST

$$\hat{c}(a) = b \hat{p}$$

- replaces the cost LCB in the LP
- fixes the early budget exhaustion

Supported by the multi-campaign ablation: pseudo-regret 72 versus 165

03 · DYNAMIC PACING

$$\rho_t = B_t / (T - t)$$

- LP cap = remaining budget / remaining rounds
- recovers spend drift around change points

Same pacing rule as the multi-campaign stage

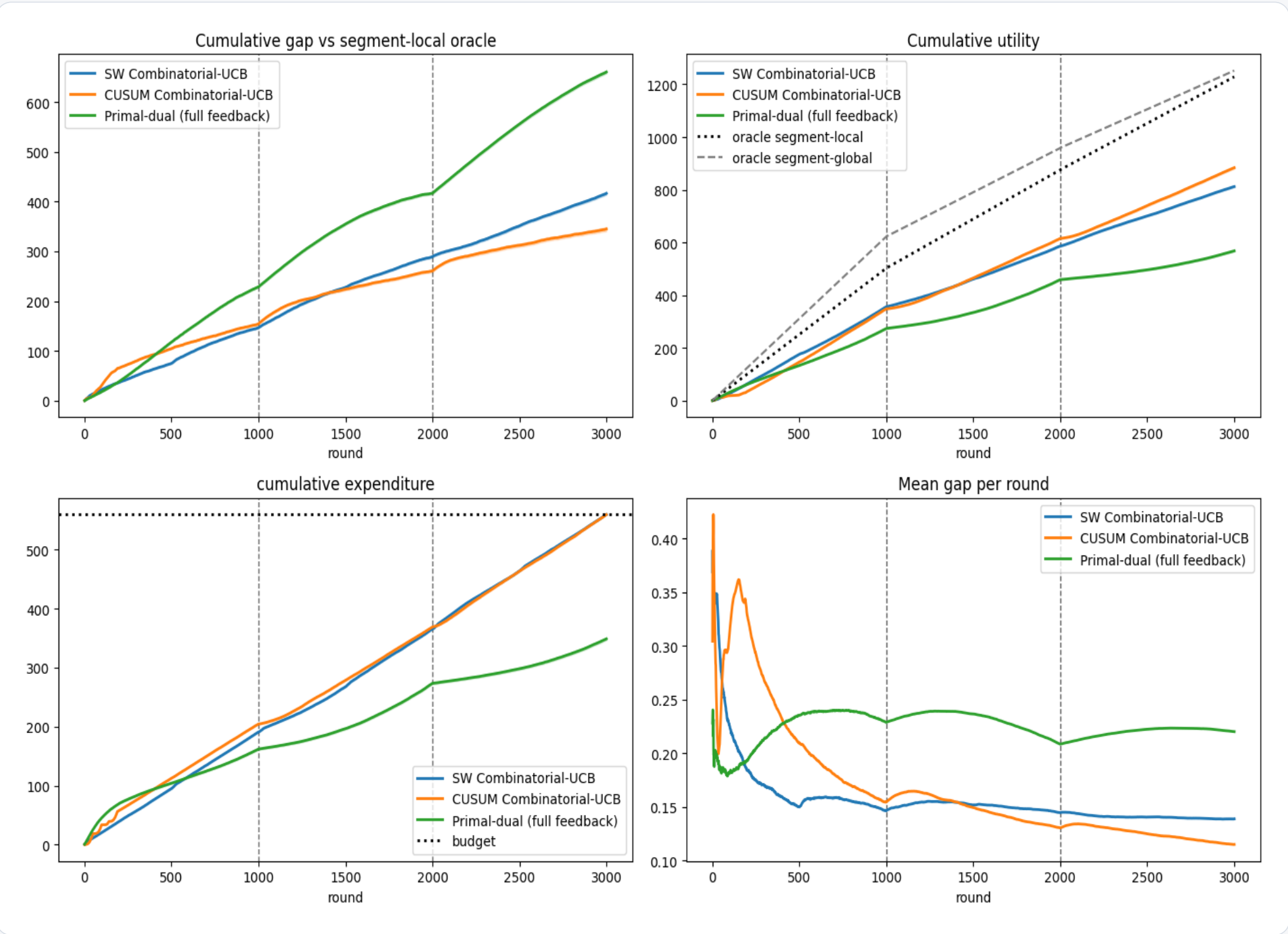
CUSUM DETECTOR UNCHANGED: only the decision layer changes through empirical cost and dynamic pacing.

PRIMAL-DUAL UPDATED: each method enters the comparison with its calibrated learning rates.

STAGE 4: EVIDENCE

CUSUM-UCB Reaches the Lowest Final Gap

30 trials with matched seeds. Every trial respects the budget.



FINAL GAP, BASELINE → IMPROVED

CUSUM-UCB 533 → 345

best method because resets concentrate near the true change points

SW-UCB 562 → 417

wider window, empirical cost, dynamic pacing

Primal-dual 716 → 660

calibrated rates, with 211 of 560 left unused

Light bar: baseline · solid bar: improved · scale up to 716.

Empirical cost and pacing extend spending to the horizon. The wider window supplies more observations per arm. CUSUM resets rise from about 18 to 32 per trial.

CUSUM-UCB Leads Under Market Drift

Average gap per round for the revised learners

0.115

CUSUM-UCB: active resets refresh statistics after each change

0.139

SW-UCB: passive forgetting pays a fixed variance cost

0.220

Primal-dual: 0.239 with the reference rates

WHAT THE COMPARISON SHOWS

- CUSUM-UCB finishes slightly ahead of SW-UCB and clearly ahead of primal-dual.
- The primal-dual benchmark is a best fixed distribution and does not track segment-specific policies.
- Calibrated rates reduce the primal-dual gap, but 211 of 560 remain unused.
- A windowed primal-dual variant improves on full memory but still trails both UCB learners.

LIMITATIONS

- CUSUM reacts slowly when observations are divided across many arms.
- A segment-global oracle can move budget across regimes, which no online learner can reproduce without future information.
- Repeated exploration after each change creates a persistent gap that grows with the number of arms.



Adaptive Budget Allocation